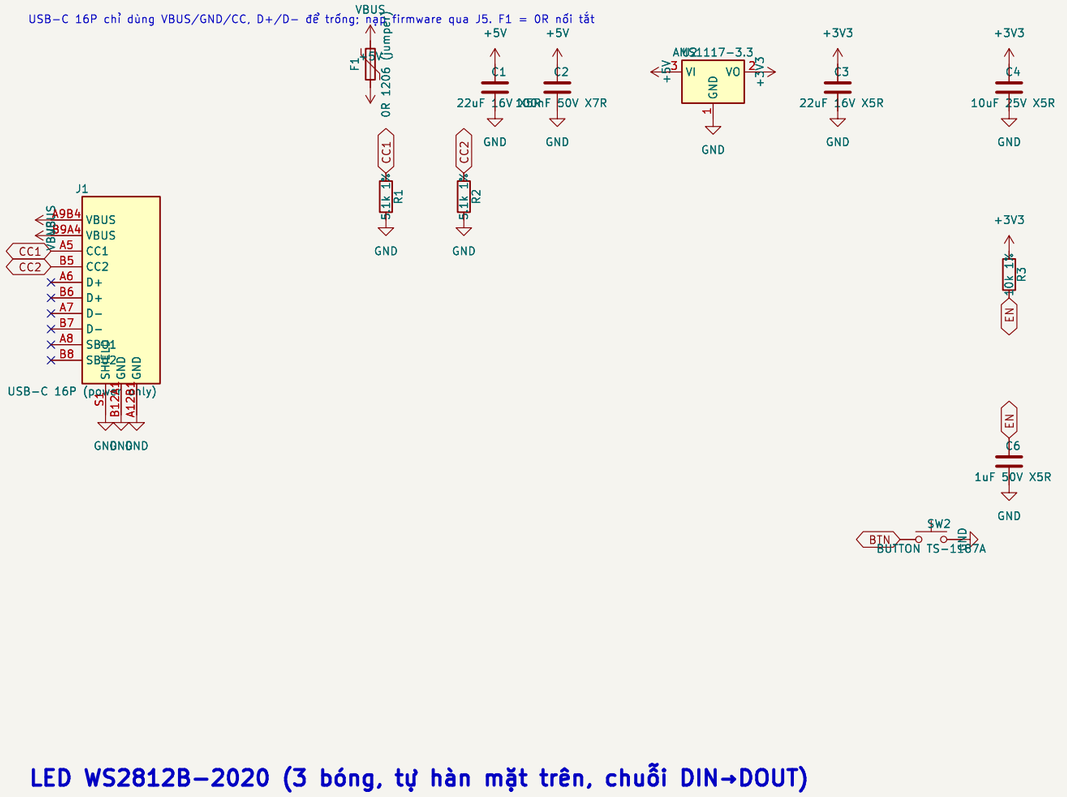
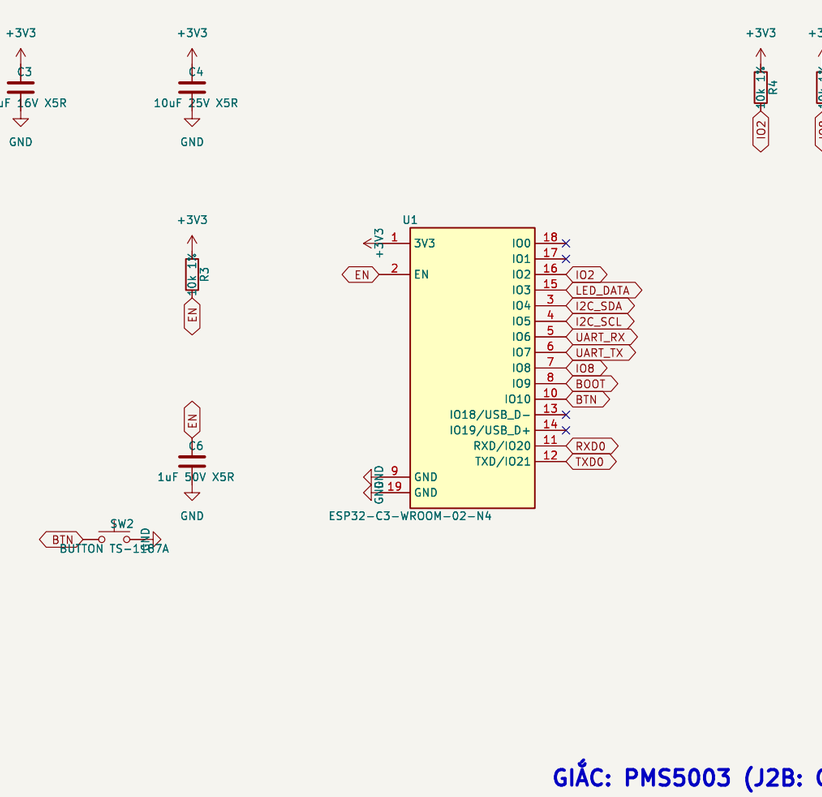
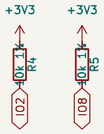
NGUỒN 5V, USB-C VÀ LDO 3.3V

USB-C 16P chỉ dùng VBUS/GND/CC, D+/D- để trống; nạp firmware qua J5. F1 = 0R nối tắt

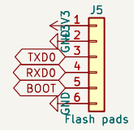


ESP32-C3-WROOM-02



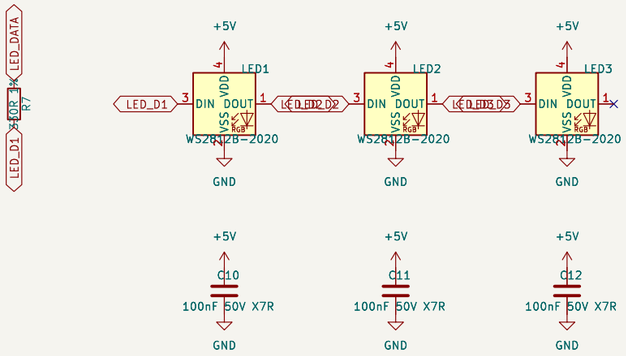


J5 - 6 PAD NẠP UART (3V3 GND TX RX IO9 GND)



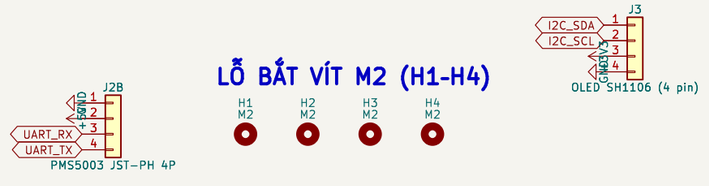
Nạp firmware: nối tắt pad 5-6 (IO9-GND) rồi cấp nguồn lại (rút/cắm USB), sau đó bỏ nối tắt. SW2 = nút ng

LED WS2812B-2020 (3 bóng, tự hàn mặt trên, chuỗi DIN→DOUT)

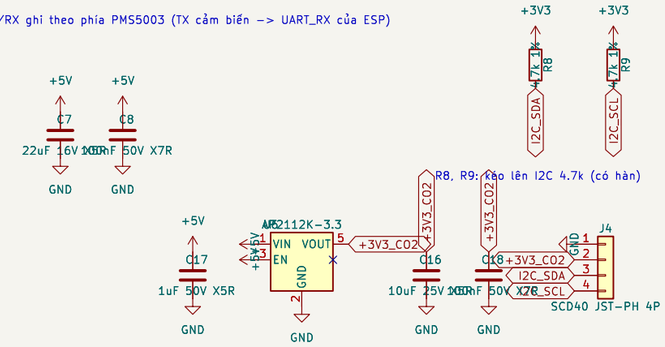


IO3 nối thẳng LED1 qua R7 330 Ω (LED nhận mức cao từ 2.8 V)

GIẮC: PMS5003 (J2B: GND VCC TX RX), OLED cắm xuyên 4 lỗ (J3)



J2B: TX/RX ghi theo phía PMS5003 (TX cảm biến -> UART_RX của ESP)



J4 - GIẮC SCD40 (JST-PH 4P) + LDO U6

PCB 51.4 x 31.4 mm, 2 lớp

Sheet: /
File: pm25.kicad_sch

Title: Cảm biến bụi PM2.5 – ESP32-C3 – LED2020

Size: A3	Date: 2026-10-28	Rev: 2.0-LED2020
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1